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MS Word or Microsoft Word is a graphical user interface (GUI) based word processing software developed by Microsoft Corporation. It is one of the most widely used software applications for creating, editing and formatting documents. It allows users to make professional documents like letters, reports, memos, resumes, etc. MS Word has many helpful tools such as grammar checker, word count, speech recognition, tables, templates, and so on.

The Home tab contains text formatting options such as font, font size, bold, italic, underline, paragraph alignment, bulleting and numbering. Insert tab is used to insert tables, images, shapes, SmartArt, charts, headers, footers, page numbers etc. Design tab provides document themes, styles, watermarks, page color and borders. Layout controls page margins, orientation, size, columns and line spacing. References is for table of contents, footnotes, citations, bibliography and index. Mailings is for mail merge, envelopes and labels. Review is for spell check, grammar check, word count

and track changes. View is for different document views like Read Mode, Print Layout, Web Layout.

To open MS Word — click on the Start button (Windows logo) on the bottom left of the screen. Scroll down the application list and find the Microsoft Office folder. Click on Word to launch the application. A splash screen will appear and then the Word home interface opens. Select Blank Document to open a new empty document.

Page Layout is found under the Layout tab in MS Word ribbon. It allows the user to control how the page looks and how content is arranged. Margins allows setting top, bottom, left and right margins — the default is 1 inch on all sides. Orientation is used to switch between Portrait (vertical) and Landscape (horizontal). Size changes paper size such as A4, Letter, Legal etc. Columns divides the page content into one, two or three columns. Indentation and Spacing adjusts line spacing and paragraph spacing.

A page break forces the remaining content to move to the next page. To insert a Page Break — place the cursor where you want the break. Go to the Insert tab and click on the Page Break option. Or simply press Ctrl + Enter from the keyboard. The content after the cursor will now begin on a new page.

To insert a picture from the computer into the Word document — go to the Insert tab from the ribbon. Click on the Pictures button. Browse and select the image from your computer files. Click Insert. The image will appear on the document. Resize or reposition the image using the drag handles.

Clip art is pre-made artwork stored in a library. In MS Word 2016 and later, Clip Art was replaced with Online Pictures. To insert — go to Insert tab, click Online Pictures, search for a keyword like computer or student, select the desired image and click Insert.

MS Word provides many built-in shapes like rectangles, circles, arrows, lines, and flowchart symbols. To insert a shape — click on Insert tab, click on the Shapes dropdown button, select any shape from the gallery, then click and drag on the

document to draw the shape. Use Shape Format tab to change fill color, outline, and effects.

Charts help to visually represent data in a document. To insert a chart in MS Word — click on the Insert tab, click on the Chart icon in the Illustrations group, select chart type such as Bar, Column, Pie or Line etc., and click OK. A small Excel spreadsheet will open to enter data. Enter your data in the spreadsheet and the chart will update in real time. Close the Excel pane and format the chart using the Chart Design tab.

MS Word is a powerful word processing tool. Through this lab we learned how to create and format a professional document including tables, headers, footers and page numbers. We also practiced inserting images, shapes, charts and applying page layout settings in the same document.

MS Excel or Microsoft Excel is a spreadsheet software developed by Microsoft Corporation. It is used to store, organize and analyze data in rows and columns. Excel is widely used in education, business, finance and scientific fields for calculations, charting and data management. It supports powerful features like formulas, functions, charts, pivot tables, conditional formatting and macros.

When you open Excel, you see a grid of rows (numbered 1, 2, 3...) and columns (labeled A, B, C...). Each box in this grid is called a cell. Each cell has a unique address like A1, B2, C3 etc. A worksheet is a single spreadsheet page and a workbook is the entire Excel file which can contain many worksheets.

Formulas in Excel always start with an equal sign (=). They perform calculations based on values in cells. Important functions used in the mark sheet are — SUM adds a range of numbers (for example =SUM(B2:F2)), AVERAGE calculates the mean value (for example =AVERAGE(B2:F2)), MAX finds the highest value in a range (for example =MAX(B2:F2)), and MIN finds the lowest value (for example =MIN(B2:F2)).

The student mark sheet was created for five students with five subjects — Nepali, English, Maths, Science and Computer. For each student, the Total Marks were calculated using the SUM function across all five subject columns. The Average Marks were computed using the AVERAGE function. The Rank was determined based on total marks. The Highest Marks and Lowest Marks for each subject were found using MAX and MIN functions at the bottom of the sheet. Conditional formatting was applied to highlight marks below 40 in red color to indicate failing grades.

For the salary sheet lab, HRA is calculated as 50% of the basic salary, TA is calculated as 10% of the basic salary. Gross Pay equals the sum of Basic, HRA and TA. A fixed deduction of Rs. 500 per employee is applied. Tax is taken at 10% of the Gross Pay. Net Pay is obtained by subtracting both Tax and Deduction from Gross Pay. All six employees had their salary components computed automatically without any manual arithmetic — changing one basic salary value would automatically update all related columns.

MS Excel is an excellent tool for financial calculations. By using cell references and formulas, we can create a complete salary management system where changing one value automatically updates all related calculations. This makes data management very fast and error-free compared to manual calculations.

Disk Operating System (DOS) is one of the oldest types of operating systems. It is a Command Line Interface (CLI) type OS where users interact with the computer by typing commands. It was widely used in the 1980s on IBM PCs. MS-DOS (Microsoft Disk Operating System) was developed by Microsoft and became the standard OS for IBM-compatible personal computers.

In modern Windows systems, we can access DOS commands through the Command Prompt (cmd.exe), which is a CLI application that simulates the old DOS environment.

To open Command Prompt — press Windows + R keys together to open the Run dialog. Type cmd in the text box and press Enter or click OK. The Command Prompt window will open with a black background. The default prompt looks like C:\Users\Mohan>

The DIR command displays the contents (files and folders) of the current directory including their size and date modified. The MKDIR command is used to create a new directory or folder in the current location — for example mkdir IIT_LAB creates a new folder named IIT_LAB. The CD command changes the current working directory — typing cd IIT_LAB moves into the IIT_LAB folder.

The COPY command copies a file from one location to another. For example copy notes.txt IIT_LAB copies the file into the IIT_LAB folder. The system displays a confirmation message that 1 file was copied successfully.

The DATE command displays and allows setting the current system date. When executed it shows the current date and asks the user to enter a new date. The TIME command similarly displays and allows setting the current system time.

The TREE command displays the entire directory structure of the current path in a visual tree format showing all folders and subfolders in a branching layout. The CLS command clears all text from the Command Prompt window and shows a fresh blank prompt. The HELP command lists all available internal DOS commands with brief descriptions of what each command does.

MS DOS commands are still relevant in modern computing through the Command Prompt. These commands allow fast file and folder management without using the graphical interface. Knowing DOS commands is an important skill for IT students as it helps in understanding how operating systems work at a fundamental level.

Microsoft PowerPoint is a presentation software application developed by Microsoft Corporation. It allows users to create slide-based presentations that can include text,

images, charts, tables, animations and transitions. PowerPoint presentations are widely used in education, business, conferences and seminars.

To create a presentation — open Microsoft PowerPoint from the Start menu and select a blank presentation or choose a theme or template. Click on the first slide and add the title and subtitle. Use the Insert tab to add new slides, pictures, tables and clip art. Apply animations from the Animations tab. Apply slide transitions from the Transitions tab. Change the color scheme from Design tab under Variants. Save the file as .pptx and run the slideshow using F5.

The five-slide Cyber Security presentation was created with a Circuit theme background. Slide 1 was the title slide with the presenter's name and institution. Slide 2 defined Cyber Security and listed key areas — Network Security, Information Security, Application Security and Operational Security. Slide 3 covered common cyber threats like Phishing, Malware, Ransomware, DDoS attacks, Man-in-the-Middle attacks and Social Engineering — a 2-column table was inserted here. Slide 4 listed best practices for online safety. Slide 5 contained the conclusion and references.

Animation effects such as Fly In on the title and Appear on bullet points were applied. Slide transitions including Wipe from left and Fade Out were used across slides. Clip art of a shield was inserted and recolored to blue on the fourth slide. The overall color scheme used was dark blue throughout all slides for consistency.

Electronic Mail (E-mail) is a method of exchanging digital messages over the internet. It allows users to send and receive messages, files, images and documents instantly to anyone in the world who has an email address. Email has largely replaced traditional postal letters for both personal and business communication. An email address has the format: username@domain.com

Gmail is a free email service developed and provided by Google LLC. It was launched on April 1, 2004. Gmail offers 15 GB of free storage space. It is accessible via a web browser at gmail.com and also through mobile apps on Android and iOS. Gmail integrates with other Google services like Google Drive, Google Meet, Google Calendar and more.

Gmail is important for fast and instant communication with anyone worldwide. It is free to use — no cost to send or receive emails. Files and documents up to 25 MB

can be attached to emails. It provides 15 GB of free cloud storage. It is very important for professional and academic communication and is used for account registration and verification on other websites.

To create a Gmail account — a working computer or smartphone is needed, along with an internet connection, a mobile phone number for verification, and a web browser such as Chrome, Firefox or Edge.

Step 1 — Open a web browser (Google Chrome or Mozilla Firefox) on your computer or laptop.

Step 2 — Go to the address bar, type www.gmail.com and press Enter.

Step 3 — The Gmail login page will appear. Click on the Create Account button below the login form.

Step 4 — A new page opens. Select the For myself option from the dropdown menu.

Step 5 — Enter your First Name and Last Name in the name fields.

Step 6 — Choose a Gmail address by typing your desired username. Gmail will suggest alternatives if the username is already taken.

Step 7 — Create a strong password and confirm it by typing it again in the next box.

Step 8 — Click Next to continue to the next step of account creation.

Step 9 — Enter your Phone Number for verification which is required for account recovery. Click Next.

Step 10 — Google will send a 6-digit OTP (One Time Password) to the phone number. Enter the OTP in the verification box and click Verify.

Step 11 — Enter additional information such as date of birth and gender. Then click Next.

Step 12 — Google will show the Privacy Policy. Scroll down and click I agree to accept the terms.

Step 13 — The Gmail account has been successfully created and you will be redirected to the Gmail inbox page.

Step 14 — You can now compose and send emails by clicking the Compose button on the left side. Inbox shows received emails. Sent shows emails that have been sent. Drafts stores unfinished emails. Spam contains junk emails. Trash holds deleted emails recoverable for 30 days. The Search Bar finds any email by keyword or sender name.

Gmail is an essential digital communication tool in today's world. Creating a Gmail account is simple and free. It gives access to email and also to the entire Google ecosystem of services. This lab helped understand the process of email account creation and the main features of the Gmail interface.